

ABOUT THE SEMINAR

Web, Crowd and Big Data Management

Changes highlighted in Yellow

Instructor

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Office: building 503, room 204

Office hours: Mondays 10:00 (by email appointment)

Website: Moodle - <https://lemida.biu.ac.il> (course website)

Requirements

Attendance on time is mandatory (late arrival = absence)

Justified absence – send relevant documents

Presenting a paper – 95% of the final grade

Participation – 5% of the final grade

- Not related to attendance!
- Discussion is the way to learn
- Ask questions and take part in activities!
- Students that cannot attend any meeting, can either:
 - (a) look at the paper/slides of at least 3 presentations and send questions
 - (b) get 100% presentation score, and present to me on Zoom

What to do

Do these ASAP!

Team up with
students on
reserve duty

1. Find a partner – presentations are in pairs
2. Choose a paper to present from the list on Moodle
Recent papers from top conferences
 - You can a paper not from the list from the conferences: SIGMOD=PACMMOD, PVLDB, ICDE, EDBT, PODS, ICDT, CIKM, The Web Conference from recent years, but it has to be at least 10 pages long and get my confirmation.
3. Choose the time for your presentation and the paper you present (Moodle)
4. Prepare presentation (aim for up to 45 mins).
5. Submit slides (Moodle)
6. Rehearse
7. Presentations are in class

Students on
reserve duty have
a priority choosing
the date

Rehearsal

- Each couple will present to the next couple ahead of the seminar to rehearse
 - Last couple will present to the first
- Scheduling is your responsibility
 - Leave enough time to incorporate the feedback
 - The schedule is on the website
- The attending couple should fill in a form (on the website)
 - They are exempt from attending the talk on the seminar
- Be helpful!
 - Suggest improvements to your friends' presentation

Ethics



- Mention the **name of the article**, **authors** and **year** in your presentation
- You may use material from the article such as figures, formulas, algorithms and citations
- **Cite other sources** you have used in the presentation
 - Example: you have used Wikipedia to explain background material
 - Example: you have used a figure from another article
 - No need to cite: you have read articles to improve your understanding, but have not used them in the presentation
 - Do not read the names in class (unless you say something about them that is relevant to the class)
 - **Good academic practice!**
- Your presentation must be **original**
 - Do not use others' slides or text
 - Do not use slides from other courses
 - Others cannot help preparing your presentation. I am here to consult.
 - You can use others' design templates and tools
- Balance the work between the partners
 - Both must show understanding (even if you do not present some part)
- **Submit your slides to Moodle** before or on your presentation day
 - They may be used in case of a concern regarding ethics

How to Read (and choose) an Academic Paper??

1. Read the abstract
2. Skim through the entire paper – now decide: choose it or not?
3. Read the introduction and the conclusion
 - Highlight important sentences
 - At this point you should understand
 - what this paper is about?
 - What is new here?
4. Fast reading of the entire paper
 - Highlight important parts
5. Now read more carefully
 - Read background material as needed
 - Summarize and highlight main points so you can go back to them later
 - You can already draft your slides at this stage

Paper must be:

- From data management top conferences (see above)
- at least 10 pages double column / 20 single column
- **Include a technical part**

A GOOD PRESENTATION

This is NOT an example for
YOUR presentations!

General Structure - Introduction

- Introduction
 - background: the general context for the paper
 - should fit the audience's knowledge
 - **Be attentive to other talks**, do not repeat – but you can remind
 - Try to be concrete
 - The problem tackled in the paper
 - A general outline of the solution, main idea or main result
 - Outline of the talk (optional) – AFTER the introduction
- Preliminaries
- Main results
- Experiments
- Summary and discussion

An interactive
part – see
below!

General Structure - Preliminaries

- Introduction
- Preliminaries
 - Similar to background, but more concrete and formal
 - Results of previous research
 - If needed, explain ideas from papers cited by your paper
- Main results
- Experiments
- Summary and discussion

General Structure – Main Results

- Introduction
- Preliminaries
- Main results
 - Can be divided into sections according to the paper structure
 - **You do not have to show everything**
 - Instead, **show some complete idea(s)**
 - Must include intuition
 - Preferably a concrete example
 - **Must include some (justified!) deep technical / formal details**
for example: formulas, algorithms, theorems and proofs, model...
- Experiments
- Summary and discussion

General Structure - Experiments

- Introduction
- Preliminaries
- Main results
- Experiments
 - If exist
 - Do not just show graphs/tables/diagrams, explain everything you show!
 - Explain the axes and the legend
 - You can elaborate on the more interesting results
 - Summarize main conclusions
- Summary and discussion

General Structure - Summary

- Introduction
- Preliminaries
- Main results
- Experiments
- Summary and discussion
 - Briefly repeat what you have presented and the main ideas
 - Discussion on future research, improvements, what else can be done?
 - Criticism on the approach/results?

Making

Good Slides



Guidelines: slides

- Less text, more visualizations
 - Images, tables, diagrams
 - Text only in short sentences, bullet-style (but not necessarily bullets)
 - Font size ≥ 14 , preferably larger (including diagrams, figures etc.)
 - The rest will be explained by you! Write notes
- Do not show anything you do not plan to explain
 - Hide slides that you want to have “just in case”
 - Do not “skim” through material – it only confuses
 - If something is not important – remove it from the presentation
- Plan the timeframe
 - A ratio of 1 minute per slide for beginner presenters, but time yourselves!

Example

Background: Location of Wikipedia Articles

Some Wikipedia articles display related location as coordinates.

The image is a screenshot of the Wikipedia article for Bar-Ilan University. It features a yellow callout box pointing to the coordinates **Coordinates: 32°4′4″N 34°50′33″E** in the article header and another yellow callout box pointing to the same coordinates in the infobox. The infobox also includes the university's name in Hebrew and its logo.

WIKIPEDIA
The Free Encyclopedia

Bar-Ilan University

Article Talk Tools

From Wikipedia, the free encyclopedia

"Bar-Ilan" redirects here. For other uses, see Bar-Ilan (disambiguation).

Bar-Ilan University (BIU, Hebrew: אוניברסיטת בר-אילן; *Universitat Bar-Ilan*) is a public research university in the Tel Aviv District city of Ramat Gan, Israel. Established in 1955, Bar Ilan is Israel's second-largest academic institution. It has about 20,000 students and 1,350 faculty members.

Bar-Ilan's mission is to "blend Jewish tradition with modern technologies and scholarship and the university endeavors to ... teach the Jewish heritage

Bar-Ilan University
אוניברסיטת בר-אילן (Hebrew)

Example

The Paper's Proposal



Wikipedia article with
unknown location



Surrogate: A Wikipedia
entity (either an article or
category) with a location
that best approximates
the desired location



Approximated location
(coordinates)
A.K.A Geolocation

Example



Choose a Surrogate

Algorithm 1 Given a set of entities E (with a weight function w) for an article a_* , return a surrogate for a_* with a confidence score

```
1: function ATTRACTORS( $E, w$ )
2:   return null if  $|E| \leq 1$ 
3:    $surrogate, score \leftarrow \text{null}, -\infty$ 
4:   for  $e \in E$  do
5:      $attraction \leftarrow 0$ 
6:     for  $e' \in E \setminus \{e\}$  do
7:        $d \leftarrow \max\{\text{dist}(\text{loc}(e), \text{loc}(e')), 1\}$ 
8:        $attraction \leftarrow attraction + \frac{w(e')}{\sqrt{d}}$ 
9:      $curr \leftarrow attraction \cdot w(e)$ 
10:    if  $curr > score$  then
11:       $surrogate, score \leftarrow e, curr$ 
12:  return  $surrogate, score$ 
```

What is the attraction score of e according to other e' ?



Tips: content

- Read and understand!

- You must be able to answer questions about the paper
 - Yes, even if they are not in your presentation
- If something is unclear, try to search and read more
- Explain upfront what you did not understand and why

Research papers are not perfect.
Sometimes they are unclear or require background knowledge. It is your role to complete necessary.
If this is impossible you can consider focusing on clearer parts.
In any case have a ready explanation why you could not complete some part.

Engage the audience

- **Part of the grade!**
- Your presentation must include an **interactive activity**. Examples:
 - A quick quiz to the audience
 - A quick game
 - Demonstrating concepts with volunteers
 - Competition between teams
 - Other?
- **Be creative!**
 - Use Kahoot, Padlet, Mentimeter, shared documents.... What else??
- Help your classmates understand
 - Asking “does everybody understand?” or “questions?” is **not** always helpful...
 - **Plan discussion intervals** by preparing “leading questions” to the audience
 - If someone asks a question, repeat it in your own words before answering
 - If there is an important and nontrivial point and nobody asks questions, **explain again!**

Challenge:

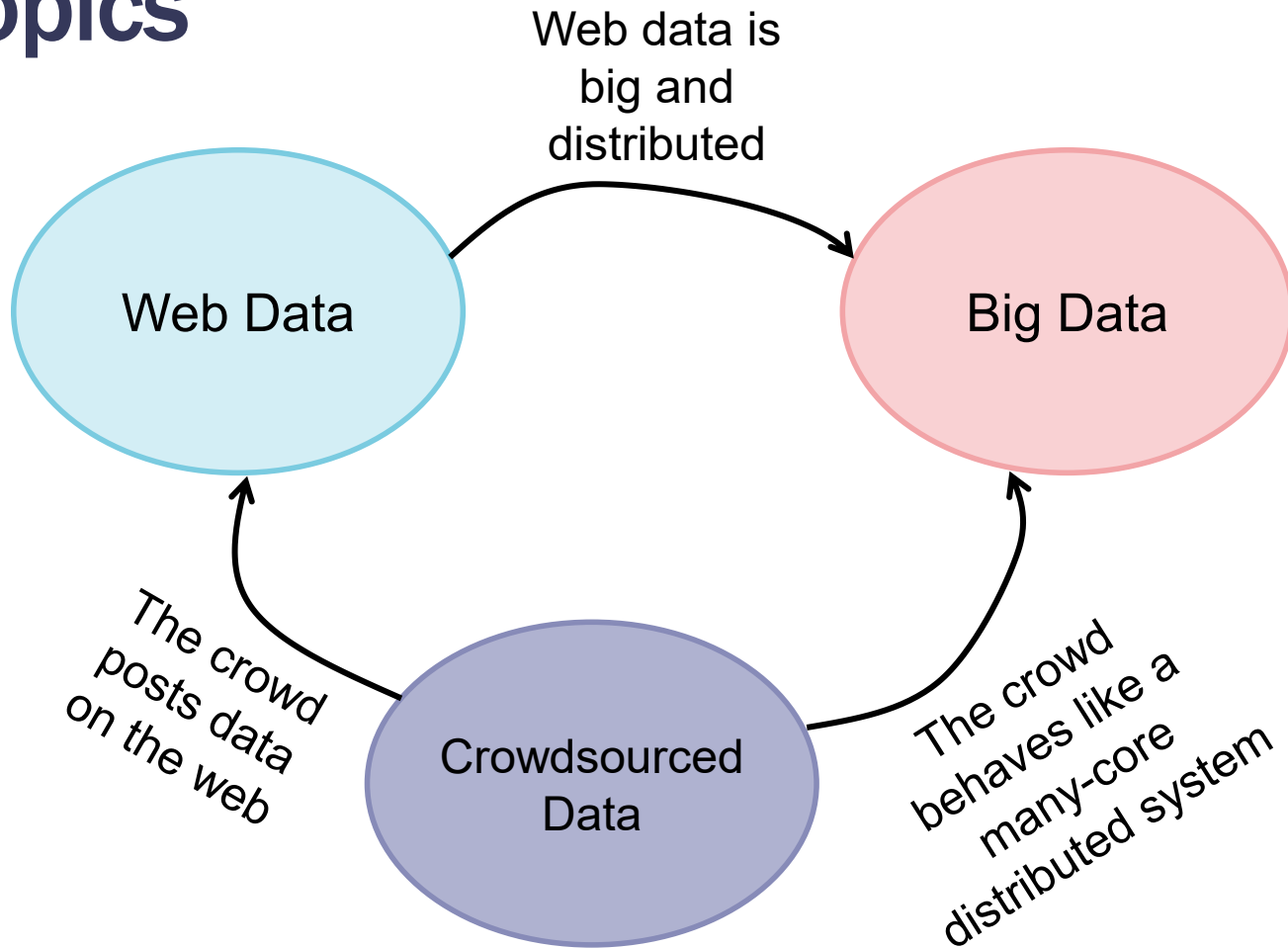
Create opportunities for participating

- Try to include as many as possible!



TOPICS OF THE SEMINAR

Topics



Web data

Characteristics

- Very distributed
- Interlinked
- Semantic web
- Social networks
- Specific formats
 - (HTML, tables)
- Varying trust
- Varying cleanliness

What to do with it?

- Clean + assess trust
- Integrate
- Explore
- Extract useful knowledge
- Analyze social patterns

Big data

Characteristics

- Very distributed
- Massive even for simple tasks (like sorting)
- Traffic is bottleneck
- Only a small % may be interesting

What to do with it?

- Parallel algorithms
- Low-traffic algorithms
- Distributed algorithms
- Prioritize
- Explore
- Extract useful insights
- Index and organize

Crowdsourced data

Characteristics

- Most of it is in people's heads
- People answer some questions more easily than computers (and vice versa!)
- Expensive and slow
- Limited attention span
- Beliefs and opinions
- Spam and malicious users
- Expertise areas of users
- Natural language

What to do with it?

- Parallel algorithms
- Tasks that require human judgment
- Settle contradictions
- Who and how many to ask?
- What to ask?
- How to incentivize?